Additional Table 4 Characteristics of studies on trehalose intervention for animal models of polyglutamine tract diseases

Studies, country	Blinding of assessor	Intervention(s), Control and setting (Intervention schedule)	Animal characteristics (Neurodegenerative diseases)	Outcome measure(s)	Clinical outcome(s)	Physiological changes	Adverse event(s)	Conclusion
Debnath et al. (2017) [21] Debnath et al.	N/R	**Intervention:** Poly(trehalose) nanoparticles.	**R6/2 transgenic mice [HD]**	Clinical N/R	N/R	Poly(trehalose) and control nanoparticles decreased level of	N/R	Trehalose-functionalized zwitterionic

(2019) [22] India		[0.4 mg/mL, injection, 10x, 2 d interval, 20 d] **Control[1]:** Control nanoparticles. (0.4 mg/mL, injection, 10x, 2 d interval, 20 d) **Control[2]:** Saline. (100 µL, injection, 10x, 2 d interval, 20 d)	Intervention: *n* = N/R Control[1]: *n* = N/R Control[2]: *n* = N/R **Background** N/R **Age (mon)** 1.4 – 2 **Gender** Female	<u>Biochemical</u> Level of plague - Huntingtin aggregates Degree of nanoparticle entering brain - Iron content		huntingtin aggregates in different brain regions of transgenic mice; greater effect was observed in poly(trehalose). Iron content increased by 3.5-fold in nanoparticle-treated brain of transgenic mice.		nanoparticles can inhibit amyloid/polyglutamine aggregation in HD through multivalent binding. Nanoscale trehalose can offer a more efficient treatment compared to molecular trehalose.
Im et al. (2013) [23] Korea	N/R	**Intervention1:** BBB-permeable trehalose derivative (TD-G6) (0.4 wt%, oral administration, until death) **Intervention[2]:** Trehalose (1.0 wt%, oral administration, until death) **Control:** Distilled water (oral administration, until death)	**R6/2 transgenic mice** **[HD]** Intervention[1]: *n* = N/R Intervention[2]: *n* = N/R Control: *n* = N/R **WT** Intervention: *n* = N/R Intervention[2]: *n* = N/R Control: *n* = N/R **Background** N/R **Age (mon)** 1 **Gender** Female	<u>Clinical</u> Clasping test - Dystonia Open field test - Locomotion and anxiety Rota-rod test - Motor coordination Lifespan <u>Biochemical</u> Level of plagues - Huntingtin aggregates	TD-G6 and trehalose improved dystonia, locomotion, and motor coordination in transgenic mice; greater effect was observed in TD-G6. TD-G6 and trehalose prolonged lifespan of transgenic mice; greater effect was observed in TD-G6. *No effect was observed in WT mice.	TD-G6 decreased level of Huntingtin aggregates in the brain of transgenic mice. TD-G6 and trehalose decreased level of Huntingtin aggregates in the liver of transgenic mice; greater effect was observed in TD-G6.	N/R	Synthetic trehalose derivative improved motor functions by reducing protein aggregation in HD with greater efficiency attributed to BBB penetration, either through stabilising partially unfolded protein or activation of autophagy.
Tanaka et al.	N/R	**Intervention[1]:**	**<u>Experiment 1</u>**	**<u>Experiment 1</u>**	**<u>Experiment 1</u>**	**<u>Experiment 1</u>**	N/R	Trehalose improved

(2004) [24] Japan	Trehalose, (2%, oral administration, daily, until death) **Control[1]:** Glucose (2%, oral administration, daily, until death) **Control[2]:** Drinking water [oral administration, daily, until death]	<u>Clinical</u> **R6/2 transgenic mice** **[HD] vs WT** Intervention[1]: *n* = 11 - Lifespan: *n* = 15 Control[2]: *n* = 10 - Lifespan: *n* = 13 **Background** N/R **Age (mon)** 1.63 – 2.57 **Gender** N/R <u>Biochemical</u> **R6/2 transgenic mice** **[HD] vs WT** Intervention[1]: *n* = 8 Control[2]: *n* = 9 **Background** N/R **Age (mon)** 1.87 – 2.8 **Gender** N/R <u>Brain atrophy</u> **R6/2 transgenic mice** **[HD]** Intervention[1]: *n* = 12	<u>Clinical</u> Rota-rod test - Motor coordination Footprint test - Motor coordination/gait and balance Lifespan <u>Biochemical</u> Level of plagues - Ubiquitin$^+$ - NeuN$^+$ - Huntingtin - β-tubulin <u>Brain atrophy</u> Ventricular dilation Average brain weight **<u>Experiment 2</u>** <u>Clinical</u> Clasping test - Dystonia Rota-rod test - Motor coordination Lifespan <u>Biochemical</u> Level of plagues - Ubiquitin$^+$	2% trehalose improved motor coordination and balance in transgenic mice. 2% trehalose prolonged lifespan of transgenic mice. **<u>Experiment 2</u>** 2% trehalose but not glucose, improved dystonia, motor coordination, and lifespan in transgenic mice.	<u>Biochemical</u> 2% trehalose reduced ubiquitin$^+$ aggregates in motor cortex, striatum, and liver of transgenic mice. 2% trehalose reduced huntingtin and β-tubulin aggregates of transgenic mice in stacking gel. <u>Brain atrophy</u> 2% trehalose reduced ventricular dilatation caused by striatum atrophy in transgenic mice. 2% trehalose increased average brain weight in transgenic mice. **<u>Experiment 2</u>** 2% trehalose, but not glucose, reduced ubiquitin$^+$ aggregates in motor cortex and striatum of transgenic mice.	motor function and survival in HD through stabilizing partially misfolded protein.

Control[2]: *n* = 11

Background
N/R

Age (mon)
2.8

Gender
N/R

Experiment 2
R6/2 transgenic mice
[HD]
Clasping test
Intervention[1]: *n* = 9
Control[1]: *n* = 14
Control[2]: *n* = 7

Rota-rod test
Intervention[1]: *n* = 11
Control[1]: *n* = 11
Control[2]: *n* = 10

Lifespan
Control[1]: *n* = 9
Control[2]: *n* = 13

Biochemical
Intervention[1]: *n* = 5
Control[1]: *n* = 6
Control[2]: *n* = 5

Background
N/R

Age (mon)
Clasping test: 1.17 – 1.63
Rota-rod test & Biochemical: 1.63

			Gender N/R					
Yang and Yu (2009) [25] China	N/R	**Intervention[1]:** C17.2 stem cell transplantation. [10 μL, injected into left lateral ventricle] + Trehalose. (2%, oral administration, daily, 10 wk) **Intervention[2]:** Trehalose. (2%, oral administration, daily, 10 wk) **Control:** Vehicle (N/R)	**R6/2 transgenic mice** **[HD]** Intervention[1]: $n \geq 3$ - Clasping: $n \geq 6$ Intervention[2]: $n \geq 3$ - Clasping: $n \geq 6$ Control: $n \geq 3$ - Clasping: $n \geq 6$ **WT** Control: $n \geq 3$ - Clasping: $n \geq 6$ **Background** B6CBAF1/J **Age (mon)** Trehalose: 0.7 Stem cell: 1.87 **Gender** Female	Clinical Clasping test - Dystonia Footprint test - Motor coordination/gait and balance Y-maze - STM, motor activity, and anxiety Rota-rod test - Motor coordination Lifespan Biochemical Level of plagues - Nestin - Ubiquitin^{+} - microtubule - β-actin Brain atrophy Striatal volume - Size of lateral ventricles	Trehalose and combined treatment with C17.2 transplant improved dystonia, motor coordination, in transgenic mice; greater effect was observed in combined treatment. Combined treatment of trehalose and C17.2 transplant improved STM, motor activity, and anxiety in transgenic mice. Trehalose prolonged lifespan of transgenic mice; 47.3% mice with combined treatment with C17.2 transplant showed further extension.	Biochemical Trehalose and combined treatment with C17.2 transplant reduced level of ubiquitin^{+} aggregates in striatum of transgenic mice by 1.65-fold and 1.23-fold. Additional effect with combined treatment with C17.2 transplant. Brain atrophy Trehalose and combined treatment with C17.2 transplant reduced size of lateral ventricles in transgenic mice; greater effect was observed in combined treatment.	N/R	A combined treatment with C17.2 stem cell improved efficacy of trehalose in HD on the basis of biochemical and behavioural criteria.
Santana et al. (2020) [26] Portugal	Yes	**Intervention[1]:** Trehalose (2%, oral administration, daily, 30 wk) **Intervention[2]:** Trehalose (2%, oral administration,	**MJD transgenic mice** **[MJD/SCA3]** Intervention[1]: $n = 7$ Control: $n = 7$ **Lentiviral-based mice** **[MJD/SCA3]** Intervention[2] (4	**Transgenic** Clinical Rota-rod test - Motor coordination Grip strength - Neuromuscular function Footprint test - Motor	**Transgenic** Trehalose improved motor coordination and gait in transgenic mice.	**Transgenic** Biochemical Trehalose decreased the diameter of mutant ataxin-3 aggregates in lobule IX of cerebellum. Brain atrophy Larger thickness of	N/R	Trehalose is a safety pharmacological strategy to counteract MJD-associated behavioural and neuropathological impairment.

	daily, 2 or 4 wk) **Control** Vehicle (Distilled water)	wk): *n* = 6; (2 wk): *n* = 6 Control (4 wk): *n* = 7; (2 wk): *n* = 6 **Background** C57BL/6 **Age (mon)** Transgenic: 0.93 Lentiviral-based: 1.87 **Gender** Transgenic: 14 Female Lentiviral-based: 25 Male	coordination/gait and balance Biochemical Level of plagues - Mutant ataxin-3 Brain atrophy Cerebellum - Volume - Layers thickness **Lentiviral-based** Clinical N/R Biochemical (2 wk) Expression of autophagy marker - LC3BII - p62 Biochemical (4 wk) Level of plagues - mutant ataxin-3 number - mutant ataxin-3 area		cerebellar layers in trehalose-treated transgenic mice. **Lentiviral-based** Biochemical (4 wk) Trehalose reduced area of aggregations although no changes in aggregate number in striatum of lentiviral-based mice.		
Chen et al. (2015) [27] Taiwan, China	N/R	**Intervention:** Trehalose (4%, oral administration, daily, 4 mon) **Control:** Vehicle (drinking water)	**SCA17 transgenic mice [SCA17]** Intervention: *n* = 13 - Cerebellum: *n* = 4 Control: *n* = 13 - Cerebellum: *n* = 4 **WT** Intervention: *n* = 13	Clinical Locomotor task - Motor coordination - Hyperactivity Rota-rod test - Motor coordination Footprint test - Motor coordination/gait and balance	Trehalose improved motor coordination and hyperactivity, and gait in transgenic mice.	Biochemical Trehalose reduced expression of $GFAP^{+}$ in cerebellum of transgenic mice. Brain atrophy Trehalose increased cerebellum weight in transgenic and WT mice.	N/R	Trehalose confers neuroprotection in SCA17 through attenuation of astrogliosis and neuron degeneration

	- Cerebellum: *n* = 4 Control: *n* = 13 - Cerebellum: *n* = 4 **Background** N/R **Age (mon)** 0.7 **Gender** 52 Male	<u>Biochemical</u> Expression of astrogliosis - GFAP^{+} Expression of chaperones - HSP70 Other molecules - MnSOD - Phospho-ERK - pp38 - β-catenin - GAD67 <u>Brain atrophy</u> Cerebellum weight

$^{+}$: Positive; BBB: blood-brain barrier; ERK: extracellular signal-regulated kinases; GAD67: glutamic acid decarboxylase 67; GFAP: glial fibrillary acidic protein; HD: Huntington's disease; LC3: microtubule-associated protein 1A/1B-light chain 3; MJD: Machado Joseph disease; MnSOD: manganese-dependent superoxide dismutase;

N/R: not reported; SCA: spinocerebellar ataxia; STM: short-term memory; WT: wild-type.